

Q1. Create an IAM policy that allows a user to download objects only from `s3://prod-logs/app1/2024/` . The user must not list the bucket, upload, delete, or access any other prefix.

Ans:

I created a custom policy named Q1-Policy that allows a user to download objects only from the

resource : `s3://prod-logs/app1/2024/` . By default the user will be denied every other permissions.

The screenshot displays the AWS IAM console interface for a custom managed policy named 'Q1-Policy'. The top section shows 'Policy details' with fields for Type (Customer managed), Creation time (February 09, 2026, 17:19 UTC+05:30), Edited time (February 09, 2026, 17:19 UTC+05:30), and ARN (arn:aws:iam::596059883112:policy/Q1-Policy). Below this, a tabbed interface shows 'Policy versions (1)' as the active tab. It indicates that each update creates a new version, with a maximum of 5 versions allowed. A table lists 'Version 1' as the 'Default' version, created 'Now'. The 'Version 1 of Q1-Policy' section shows the JSON policy document, which allows the 's3:GetObject' action on the resource 'arn:aws:s3:::prod-logs/app1/2024/'. A 'Copy JSON' button is available next to the policy text.

```
1- [{
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Sid": "Statement1",
6-       "Effect": "Allow",
7-       "Action": [
8-         "s3:GetObject"
9-       ],
10-      "Resource": [
11-        "arn:aws:s3:::prod-logs/app1/2024/"
12-      ]
13-    }
14-  ]
15- }]
```

Q2. Create an IAM role that allows start and stop actions only on EC2 instances tagged Environment=Dev. All other EC2 actions and instances must be denied, even if additional permissions exist.

Ans:

To create an IAM role with respective function/actions , first I created a policy named Q2-Policy which allows start and stop actions only on the instances tagged Environment = Dev using the stringequals in the policy and denying the instances the permissions which don't have the tag as Environment = Dev.

Creating policy

Customer managed February 09, 2026, 21:54 (UTC+05:30) February 09, 2026, 21:54 (UTC+05:30) am:aws:iam::596059883112:policy/Q2Policy

Permissions Entities attached Tags Policy versions (1) Last Accessed

This policy defines some actions, resources, or conditions that do not provide permissions. To grant access, policies must have an action that has an applicable resource or condition. For details, choose **Show remaining**. [Learn more](#)

Permissions defined in this policy [Info](#) [Copy](#) [Edit](#) [Summary](#) [JSON](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

```

1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Sid": "Statement1",
6-       "Effect": "Allow",
7-       "Action": [
8-         "ec2:StartInstances",
9-         "ec2:StopInstances"
10-      ],
11-       "Resource": [
12-         "arn:aws:ec2:*:*:instance/*"
13-      ],
14-       "Condition": {
15-         "StringEqualsIfExists": {
16-           "aws:ResourceTag/Environment": "Dev"
17-         }
18-       }
19-     },
20-     {
21-       "Sid": "Statement2",
22-       "Effect": "Deny",
23-       "Action": [
24-         "ec2:StartInstances",
25-         "ec2:StopInstances"
26-      ],
27-       "Resource": [
28-         "arn:aws:ec2:*:*:instance/*"
29-      ],
30-       "Condition": {
31-         "StringNotEquals": {
32-           "aws:ResourceTag/Environment": "Dev"
33-         }
34-       }
35-     },
36-     {
37-       "Sid": "Statement3",
38-       "Effect": "Deny",
39-       "NotAction": [
40-         "ec2:StartInstances",
41-         "ec2:StopInstances"
42-      ],
43-       "Resource": [
44-         "arn:aws:ec2:*:*:instance/*"
45-      ],
46-       "Condition": {
47-         "StringNotEquals": {
48-           "aws:ResourceTag/Environment": "Dev"
49-         }
50-       }
51-     }
52-   ]
53- }

```

The policy allows only the instances tagged with "Dev" to start and stop instances and deny all other EC2 actions and instances are denied.

Now attaching it to the role.

Q2 [Info](#) [Delete](#)

Allows EC2 instances to call AWS services on your behalf.

Summary [Edit](#)

Creation date February 09, 2026, 21:59 (UTC+05:30)	ARN am:aws:iam::596059883112:role/Q2	Instance profile ARN arn:aws:iam::596059883112:instance-profile/Q2
Last activity -	Maximum session duration 1 hour	

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (1) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
Q2Policy	Customer managed	1

► **Permissions boundary** (not set)

The above IAM role is attached with the policy that does the said task.

Q3. Create an IAM policy that denies `ec2:TerminateInstances` for all EC2 instances except those tagged `Owner=AdminTeam`.

Ans:

The policy is created globally to deny `ec2:TerminateInstances` permission for all the ec2

instances except those tagged with Owner= "AdminTeam".

Policy details

Type Customer managed	Creation time February 09, 2026, 22:21 (UTC+05:30)	Edited time February 09, 2026, 22:21 (UTC+05:30)	ARN arn:aws:iam::596059883112:policy/Q3Policy
--------------------------	---	---	--

Policy versions (1)

Each time you update a policy, you create a new version. You can have up to 5 versions of a customer managed policy.

Set as default Delete

< 1 > ⚙

Policy version ▾ Creation time ▾

Version 1 **Default** Now

Version 1 of Q3Policy [Copy JSON](#)

```
1- [{"Version": "2012-10-17",
2-   "Statement": [
3-     {
4-       "Sid": "Statement1",
5-       "Effect": "Deny",
6-       "Action": [
7-         "ec2:TerminateInstances"
8-       ],
9-       "Resource": [
10-        "arn:aws:ec2:*:*:instance/*"
11-      ],
12-       "Condition": {
13-         "StringNotEquals": {
14-           "aws:ResourceTag/Owner": "AdminTeam"
15-         }
16-       }
17-     }
18-   ]
19- }]
```

Q4. Configure cross-account access where an IAM role in Account A can read objects from an S3 bucket in Account B using sts:AssumeRole. Implement the trust policy and permission policies without using access keys.

Ans:

I am now account B that is my bucket will be read by another account.

So First I create a role and add sts:AssumeRole policy to it and added the account of friend's id.

CrossRole Info

Cross

Summary

Creation date February 11, 2026, 16:33 (UTC+05:30)	ARN arn:aws:iam::596059883112:role/CrossRole	Link to switch roles in console https://signin.aws.amazon.com/switchrole?roleName=CrossRole&account=596059883112
Last activity -	Maximum session duration 1 hour	

Permissions **Trust relationships** Tags Last Accessed Revoke sessions

Trusted entities [Edit trust policy](#)

Entities that can assume this role under specified conditions.

```
1- [{"Version": "2012-10-17",
2-   "Statement": [
3-     {
4-       "Effect": "Allow",
5-       "Principal": {
6-         "AWS": "arn:aws:iam::842533608113:root"
7-       },
8-       "Action": "sts:AssumeRole",
9-       "Condition": {}
10-     }
11-   ]
12- }]
```

Now the configurations in the another account of friend was added.

First created role and attached the policy to it using trust policy edited the role and added the bukctet arn and the permissions needed.

ec2CrossRole

Allows EC2 Instances to call AWS services on your behalf.

Summary

Creation date: February 11, 2026, 16:36 (UTC+05:30)

ARN: arn:aws:iam::842533680113:role/ec2CrossRole

Instance profile ARN: arn:aws:iam::842533680113:instance-profile/ec2CrossRole

Last activity: -

Maximum session duration: 1 hour

Permissions policies (1)

You can attach up to 10 managed policies.

Permissions boundary (not set)

CpPolicy

Policy details

Type: Customer managed

Creation time: February 11, 2026, 16:38 (UTC+05:30)

Edited time: February 11, 2026, 16:38 (UTC+05:30)

ARN: arn:aws:iam::842533680113:policy/CpPolicy

Permissions defined in this policy

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": "sts:AssumeRole",
7-       "Resource": "arn:aws:iam::596059883112:role/CrossRole"
8-     }
9-   ]
10- }
```

aws

```
ubuntu@ip-172-31-19-23:~$ aws sts get-caller-identity
{
  "UserId": "AROAA1KXLPFY262L74BKO:1-079040329ebd308c2",
  "Account": "842533680113",
  "Arn": "arn:aws:sts::842533680113:assumed-role/ec2CrossRole/1-079040329ebd308c2"
}

ubuntu@ip-172-31-19-23:~$ aws sts assume-role \
> --role-arn arn:aws:iam::596059883112:role/CrossRole \
> --role-session-name TestSession
{
  "Credentials": {
    "AccessKeyId": "ASIAYVR7QAZUOHWSGFF",
    "SecretAccessKey": "4Pmyg8hayEzftXbHJ7ocv/8R9sk+6WNME4ESq0l",
    "SessionToken": "IQoJb3Jp221uX2VjEPT//////////wEaCkVzLWVhc3QtMSJlMEYlcjQ0BmK121pup2alnoUDDel/99joIdsscxB93f1oMNRKtQ1hAJ7wc+GXVYUW+mOoAswuG9zAF6Yw6vJnh2113Kn4AW14KqECCLz//////////wEQABoMNTK2MDU5ODg2MTY5IgxbrJ+9IKch2kfbubQ9QGtFsAryem/OxEgkGIAUJ2/CdJglaeHyZfPF7c56qXenUkQzHiy9Uw1KjnUHQMGVAbDzzXZ/nMTr8keF79c1A/Xw/iveKQtVsZe1AZ150weF0zzpMRyq/Hq1w4+HMU59vfkNUjEzaJp91D11ZnFeshl3yxPtb5S8CxbYjK3ZTUDbw9VrIdS011168mNltEHKwon3lUos1ie9QIhE88zCnc/Yx7qK2uQIQ2w2mjvWTEdLvfslqe948uRAHbNM9akUwLu9kdxleloCUISE/XoWVUn7kmlsRU1YwBBbKLCkNfac5RXJxreuBsegaJotoupN1TDDybHMBjqcAQEZHRBpZzQuOxg+r3HT66v6WDJmUTs3BTWYBUevRyc7vdtntBnXUHAtaeoLU918/u0zDUZJt3n53YW/XcHgtXY1UyeB1lPkgI/OAbVV2BEZiaf/bMRFARTl1hGS1+UN4NucwflNYC3l1xpCmmnoFCX62X2nWZ1WTU4AzoUoVlotJAX44UOT+m79FByG1gE10vrJ5CGKgrg==",
    "Expiration": "2026-02-11T12:15:15+00:00"
  },
  "AssumedRoleUser": {
    "AssumedRoleId": "AROAYVR7QAZUD4QXJG2JC:TestSession",
    "Arn": "arn:aws:sts::596059883112:assumed-role/CrossRole/TestSession"
  }
}

ubuntu@ip-172-31-19-23:~$ export AWS_ACCESS_KEY_ID=ASIAYVR7QAZUOHWSGFF
ubuntu@ip-172-31-19-23:~$ export AWS_SECRET_ACCESS_KEY=4Pmyg8hayEzftXbHJ7ocv/8R9sk+6WNME4ESq0l
ubuntu@ip-172-31-19-23:~$ export AWS_SESSION_TOKEN=IQoJb3Jp221uX2VjEPT//////////wEaCkVzLWVhc3QtMSJlMEYlcjQ0BmK121pup2alnoUDDel/99joIdsscxB93f1oMNRKtQ1hAJ7wc+GXVYUW+mOoAswuG9zAF6Yw6vJnh2113Kn4AW14KqECCLz//////////wEQABoMNTK2MDU5ODg2MTY5IgxbrJ+9IKch2kfbubQ9QGtFsAryem/OxEgkGIAUJ2/CdJglaeHyZfPF7c56qXenUkQzHiy9Uw1KjnUHQMGVAbDzzXZ/nMTr8keF79c1A/Xw/iveKQtVsZe1AZ150weF0zzpMRyq/Hq1w4+HMU59vfkNUjEzaJp91D11ZnFeshl3yxPtb5S8CxbYjK3ZTUDbw9VrIdS011168mNltEHKwon3lUos1ie9QIhE88zCnc/Yx7qK2uQIQ2w2mjvWTEdLvfslqe948uRAHbNM9akUwLu9kdxleloCUISE/XoWVUn7kmlsRU1YwBBbKLCkNfac5RXJxreuBsegaJotoupN1TDDybHMBjqcAQEZHRBpZzQuOxg+r3HT66v6WDJmUTs3BTWYBUevRyc7vdtntBnXUHAtaeoLU918/u0zDUZJt3n53YW/XcHgtXY1UyeB1lPkgI/OAbVV2BEZiaf/bMRFARTl1hGS1+UN4NucwflNYC3l1xpCmmnoFCX62X2nWZ1WTU4AzoUoVlotJAX44UOT+m79FByG1gE10vrJ5CGKgrg==
ubuntu@ip-172-31-19-23:~$ aws s3 ls s3://aws-s3-demo-bucket-q7
PRE Unsaved/
2026-02-11 04:22:37 1205 about.html
2026-02-11 04:34:32 872 index.html
```

The account A was able to read and list my buckets object . The screenshots show the working of the roles.

Q5. Create an IAM policy that allows read-only access to:

- One specific EC2 instance
- One specific RDS DB instance
- All other EC2 and RDS resources must be denied.

Ans:

I first created the ec2-instance to get specific instance id to attach in the policy to get read-only access and similarly the RDS DB instance as well . Then Using the effect : Allow and action which are necessary for the question. The specify explicit deny to all other RDS DB instance and the Ec2 instance as well.

Policy details

Type Customer managed	Creation time February 09, 2026, 22:48 (UTC+05:30)	Edited time February 09, 2026, 22:48 (UTC+05:30)	ARN arn:aws:iam::596059883112:policy/Q5Policy
--------------------------	---	---	--

Permissions | Entities attached | Tags | **Policy versions (1)** | Last Accessed

Versions of this policy (1) [Info](#)

Each time you update a policy, you create a new version. You can have up to 5 versions of a customer managed policy.

< 1 > ⚙

Policy version | Creation time

Version 1 **Default** Now

Version 1 of Q5Policy [Copy JSON](#)

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Effect": "Allow",
7       "Action": [
8         "ec2:DescribeInstanceStatus",
9         "ec2:DescribeInstances"
10      ],
11      "Resource": [
12        "arn:aws:ec2:eu-north-1:596059883112:instance/i-0fde61961876b9cf8"
13      ]
14    },
15    {
16      "Sid": "Statement2",
17      "Effect": "Allow",
18      "Action": [
19        "rds:DescribeDBInstances"
20      ],
21      "Resource": [
22        "arn:aws:rds:eu-north-1:596059883112:db:database-1"
23      ]
24    },
25    {
26      "Sid": "Statement3",
27      "Effect": "Deny",
28      "NotAction": [
29        "rds:DescribeDBInstances",
30        "ec2:DescribeInstances",
31        "ec2:DescribeInstanceStatus"
32      ],
33      "Resource": [
34        "arn:aws:rds:*:*:db:*",
35        "arn:aws:ec2:*:*:instance/*"
36      ]
37    }
38  ]
39 }
```

The not action in the policy works as the action which says that deny every action except the one we listed above.

Q6. Create an IAM role that allows:

Viewing RDS DB instances

Connecting to only one specified RDS instance

Denies access to all other RDS instances

Ans:

First I created the policy to allow viewing RDS DB instances by specifying action : DescribeDBInstances and then used another statement to add connection to only a specific RDS instance using the Specific ARN , and then added another statement in the policy that restrict the access to all the other RDS instances i.e denies access to all the other DB instances.

Policy details

Type

Customer managed

Creation time

February 09, 2026, 23:37 (UTC+05:30)

Edited time

February 09, 2026, 23:37 (UTC+05:30)

ARN

arn:aws:iam::596059883112:policy/Q6Policy

Permissions

Entities attached

Tags

Policy versions (1)

Last Accessed

Versions of this policy (1) Info

Set as default

Delete

Each time you update a policy, you create a new version. You can have up to 5 versions of a customer managed policy.

Policy version

Creation time

Version 1

Default

Now

Version 1 of Q6Policy

Copy JSON

```

1- [
2-   {
3-     "Version": "2012-10-17",
4-     "Statement": [
5-       {
6-         "Sid": "Statement1",
7-         "Effect": "Allow",
8-         "Action": [
9-           "rds:DescribeDBInstances",
10-          "rds:ListTagsForResource"
11-        ],
12-         "Resource": [
13-           "*"
14-        ],
15-       },
16-       {
17-         "Sid": "Statement2",
18-         "Effect": "Allow",
19-         "Action": [
20-           "rds-db:connect"
21-        ],
22-         "Resource": [
23-           "arn:aws:rds-db:eu-north-1:596059883112:dbuser:db-VTU36NKEIYYHKKL02FGXNOUL24/admin"
24-        ],
25-       },
26-       {
27-         "Sid": "Statement3",
28-         "Effect": "Deny",
29-         "NotAction": [
30-           "rds:DescribeDBInstances",
31-           "rds:ListTagsForResource",
32-           "rds-db:connect"
33-        ],
34-         "Resource": [
35-           "arn:aws:rds:*:*:db:*"
36-        ],
37-       }
38-     ]
39-   }
40- ]

```

Now creating an IAM role and attaching it with the policy.

RDS-ACCESS-ROLE Info

Delete

RDS access role.

Summary

Edit

Creation date

February 09, 2026, 23:39 (UTC+05:30)

ARN

arn:aws:iam::596059883112:role/RDS-ACCESS-ROLE

Instance profile ARN

arn:aws:iam::596059883112:instance-profile/RDS-ACCESS-ROLE

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1/1) Info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

Policy name

Type

Attached entities

Q6Policy

Customer managed

1

Q6Policy

Copy JSON

Edit

```

1- [
2-   {
3-     "Version": "2012-10-17",
4-     "Statement": [
5-       {
6-         "Sid": "Statement1",
7-         "Effect": "Allow",
8-         "Action": [
9-           "rds:DescribeDBInstances",
10-          "rds:ListTagsForResource"
11-        ],
12-         "Resource": [
13-           "*"
14-        ],
15-       }
16-     ]
17-   }
18- ]

```

The above IAM role has been attached to the policy .

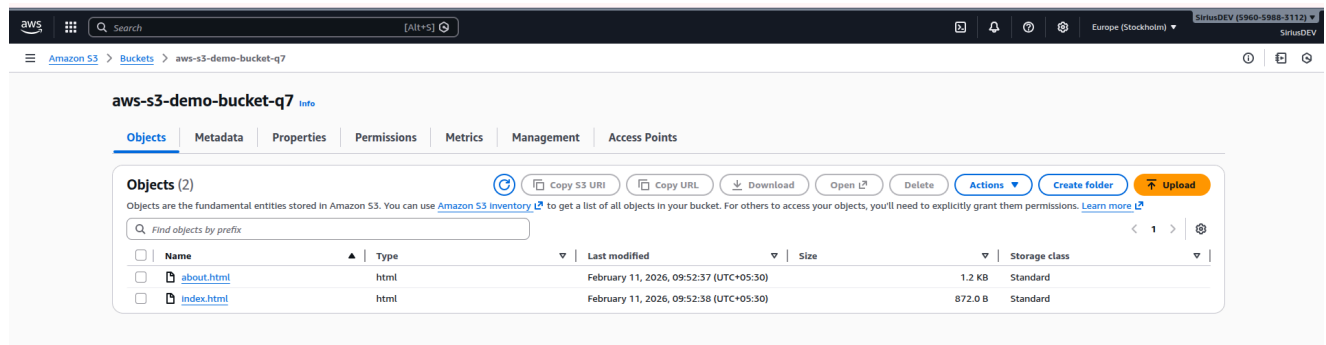
Q7. Enable CloudTrail in a single AWS account and configure it to log:

- Management events
- S3 data events

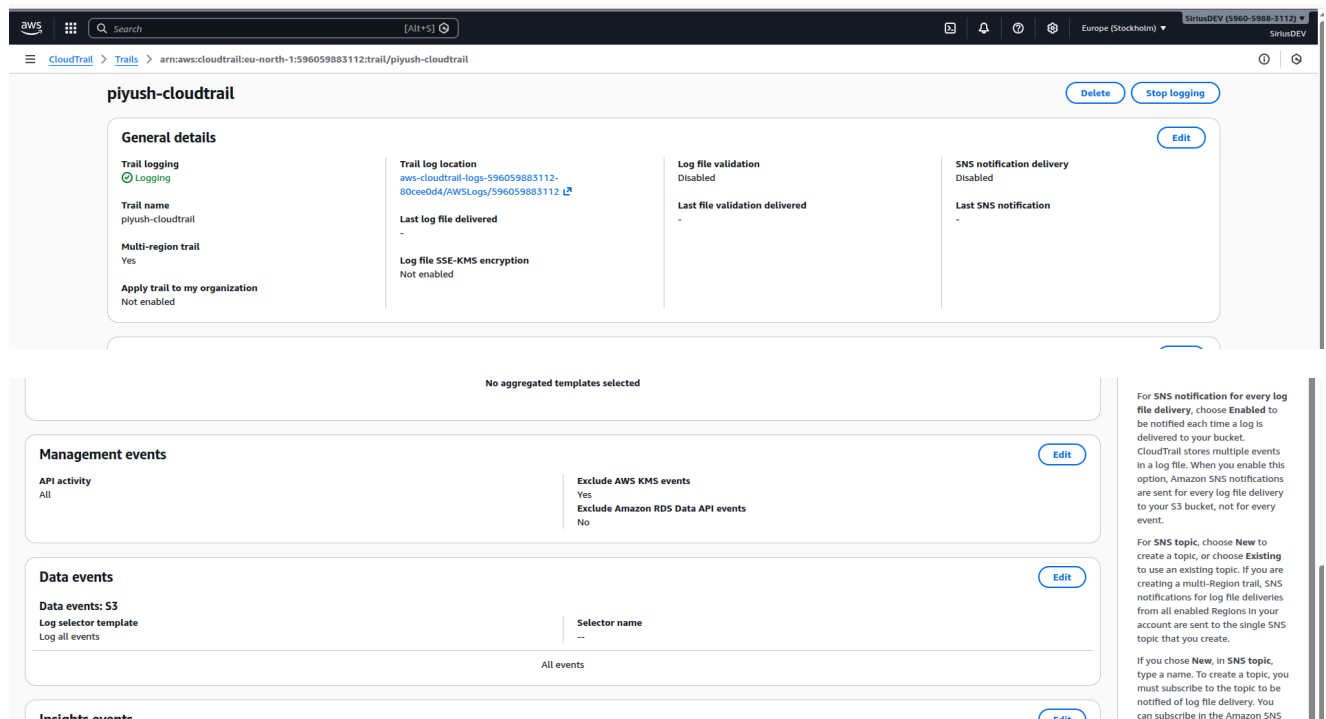
Store the logs in S3 with log file validation enabled.

Ans:

First I created my own bucket and added some custom html pages to serve.



Now creating a trail using the CloudTrail and then configuring it to view the logs.



It created a s3 bucket on my behalf which will store all the logs relating to management events and s3 data events. Now checking the logs.

loudtrail-logs-596059883112-80cee0d4 > AWSLogs/ > 596059883112/ > CloudTrail/ > eu-north-1/ > 2026/ > 02/ > 11/

11/ Copy S3 URI

Objects Properties

Objects (5) Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	596059883112_CloudTrail_eu-north-1_20260211T0435Z_bNIPAB89gbQxOUoUA.json.gz	gz	February 11, 2026, 10:00:57 (UTC+05:30)	8.1 KB	Standard
☐	596059883112_CloudTrail_eu-north-1_20260211T0440Z_9TKFPlv2KufE3yUxU.json.gz	gz	February 11, 2026, 10:08:14 (UTC+05:30)	9.7 KB	Standard
☐	596059883112_CloudTrail_eu-north-1_20260211T0440Z_JDgCNpVat7zGPfe5.json.gz	gz	February 11, 2026, 10:10:41 (UTC+05:30)	8.0 KB	Standard
☐	596059883112_CloudTrail_eu-north-1_20260211T0440Z_RwqJ5JtXehJ2dXnm.json.gz	gz	February 11, 2026, 10:08:31 (UTC+05:30)	3.0 KB	Standard
☐	596059883112_CloudTrail_eu-north-1_20260211T0440Z_y5e24CJE41F0dCjB.jsn.gz	gz	February 11, 2026, 10:05:48 (UTC+05:30)	13.1 KB	Standard

Pretty-print ☒

```
{
  "Records": [
    {
      "eventVersion": "1.11",
      "userIdentity": {
        "type": "AssumedRole",
        "principalId": "AROAYVR7QAZUG4KI0FFDZ:AutoScaling",
        "arn": "arn:aws:sts:596059883112:assumed-role/AWSServiceRoleForAutoScaling/AutoScaling",
        "accountId": "596059883112",
        "sessionContext": {
          "sessionIssuer": {
            "type": "Role",
            "principalId": "AROAYVR7QAZUG4KI0FFDZ",
            "arn": "arn:aws:iam:596059883112:role/aws-service-role/autoscaling.amazonaws.com/AWSServiceRoleForAutoScaling",
            "accountId": "596059883112",
            "userName": "AWSServiceRoleForAutoScaling"
          },
          "attributes": {
            "creationDate": "2026-02-11T04:15:06Z",
            "mfaAuthenticated": "false"
          }
        }
      },
      "invokedBy": "autoscaling.amazonaws.com",
      "eventTime": "2026-02-11T04:27:04Z",
      "eventSource": "ec2.amazonaws.com",
      "eventName": "DescribeInstanceStatus",
      "awsRegion": "eu-north-1",
      "sourceIPAddress": "autoscaling.amazonaws.com",
      "userAgent": "autoscaling.amazonaws.com",
      "requestParameters": {
        "instancesSet": {
          "items": [
            {
              "instanceId": "i-0df4b56bba33f8eal"
            }
          ]
        }
      },
      "filterSet": {
        "includeAllInstances": true,
        "ignoreManagedInstancesVisibilityConfiguration": false
      },
      "responseElements": null,
      "requestID": "ad32c807-7273-449d-a35f-a74ac233ed30",
      "eventID": "cd3d175e-f8f0-4660-a080-84dddb782a9b",
      "readOnly": true,
      "eventType": "AwsApiCall",
      "managementEvent": true,
      "recipientAccountId": "596059883112",
      "eventCategory": "Management"
    },
    {
      "eventVersion": "1.11",
      "userIdentity": {
        "type": "AssumedRole",
        "principalId": "AROAYVR7QAZUG4KI0FFDZ:AutoScaling",
        "arn": "arn:aws:sts:596059883112:assumed-role/AWSServiceRoleForAutoScaling/AutoScaling",
        "accountId": "596059883112",
        "accessKeyId": "ASIAVVR7QAZUASVH5T67",
        "sessionContext": {

```

Q8. Create an Athena table for CloudTrail logs stored in S3 and write a query to list:

- All TerminateInstances and DeleteDBInstance API calls
- Event time, IAM principal, and source IP address

*Ans:**

First on athena I created a query result location bucket where the results of queries will be stored . The bucket I am using is from the previous question only.

Amazon Athena > Query editor

Settings successfully updated.

Query in Amazon SageMaker Unified Studio Set up X

Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment.

Editor Recent queries Saved queries Query settings Workgroup primary

Query settings

Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.

Query result encryption			
Query result location s3://aws-s3-demo-bucket-q/	Encrypt query results -	Expected bucket owner 596059883112	Assign bucket owner full control over query results Turned on

Manage

Now creating the cloudtrail database for querying in athena.

The screenshot shows the AWS Athena console interface. At the top, there's a 'Query 1' header. Below it, the SQL editor contains the following code:

```
1 CREATE DATABASE cloudtrail_logs;
```

The status bar at the bottom indicates 'SQL Ln 2, Col 1'. On the right side, there are buttons for 'Run again', 'Explain', 'Cancel', 'Clear', and 'Create'. A 'Reuse query results' toggle is also visible, set to 'up to 60 minutes ago'.

Created the table for cloudtrail logs

The screenshot shows the AWS Athena console interface. The SQL editor contains the following code:

```
1 CREATE EXTERNAL TABLE cloudtrail_logs.cloudtrail_events (  
2   eventVersion STRING,  
3   userIdentity STRUCT<  
4     type: STRING,  
5     principalId: STRING,  
6     arn: STRING,  
7     accountId: STRING,  
8     userName: STRING  
9   >,  
10  eventTime STRING,  
11  eventSource STRING,  
12  eventName STRING,  
13  awsRegion STRING,  
14  sourceIPAddress STRING,  
15  userAgent STRING,  
16  eventName STRING,  
17  awsRegion STRING,  
18  sourceIPAddress STRING,  
19  userAgent STRING,  
20  requestParameters STRING,  
21  responseElements STRING,  
22  additionalEventData STRING,  
23  eventID STRING,  
24  eventType STRING,  
25  recipientAccountId STRING  
26 )  
27 PARTITIONED BY ([region STRING, year STRING, month STRING, day STRING])  
28 ROW FORMAT SERDE 'org.openx.data.jsonserde.JsonSerDe'  
29 LOCATION 's3://aws-cloudtrail-logs-596059883112-80cee0d4/AWSLogs/596059883112/CloudTrail/';
```

The status bar at the bottom indicates 'SQL Ln 23, Col 70'. On the right side, there are buttons for 'Run again', 'Explain', 'Cancel', 'Clear', and 'Create'. A 'Reuse query results' toggle is also visible, set to 'up to 60 minutes ago'.

Now querying on the table created to get the desired answers:

Query to list all terminate instances and delete instances api calls

The screenshot shows the AWS Athena console interface. The SQL editor contains the following code:

```
1 SELECT  
2   eventTime,  
3   userIdentity.arn AS iam_principal,  
4   sourceIPAddress,  
5   eventName  
6 FROM cloudtrail_logs.cloudtrail_logs_aws_cloudtrail_logs_596059883112_80cee0d4  
7 WHERE eventName IN ('TerminateInstances', 'DeleteDBInstance')  
8 ORDER BY eventTime DESC;
```

The status bar at the bottom indicates 'SQL Ln 9, Col 1'. Below the editor, there are tabs for 'Query results' and 'Query stats'. The 'Query results' tab is active, showing a 'Completed' status with a green checkmark. The query execution details are: 'Time in queue: 62 ms', 'Run time: 667 ms', and 'Data scanned: 265.61 KB'. There are buttons for 'Copy' and 'Download results CSV'. Below the details, there's a search bar and a table with 5 results. The table has columns: '#', 'eventTime', 'iam_principal', 'sourceIPAddress', and 'eventName'.

#	eventTime	iam_principal	sourceIPAddress	eventName
1	2026-02-11T05:14:38Z	arn:aws:iam::596059883112:root	106.210.36.40	TerminatelInstances
2	2026-02-11T05:12:48Z	arn:aws:sts::596059883112:assumed-role/AWSServiceRoleForAutoScaling/AutoScaling	autoscaling.amazonaws.com	TerminatelInstances
3	2026-02-11T05:09:13Z	arn:aws:sts::596059883112:assumed-role/AWSServiceRoleForAutoScaling/AutoScaling	autoscaling.amazonaws.com	TerminatelInstances
4	2026-02-11T05:07:10Z	arn:aws:iam::596059883112:root	106.210.36.40	TerminatelInstances
5	2026-02-11T05:06:57Z	arn:aws:iam::596059883112:root	106.210.36.40	TerminatelInstances

Q9. You are working as a DevOps Engineer in an organization that runs production workloads on Amazon EC2. The application running on the EC2 instance needs secure access to multiple AWS services. The security team has mandated the use of IAM roles (instance profiles) and resource-based policies to enforce least privilege. Your objective is to design, implement, and validate a secure access model using both identity-based and resource-based policies.

- Amazon S3 – to read application configuration files and write application logs.

- AWS Secrets Manager – to securely retrieve database credentials.

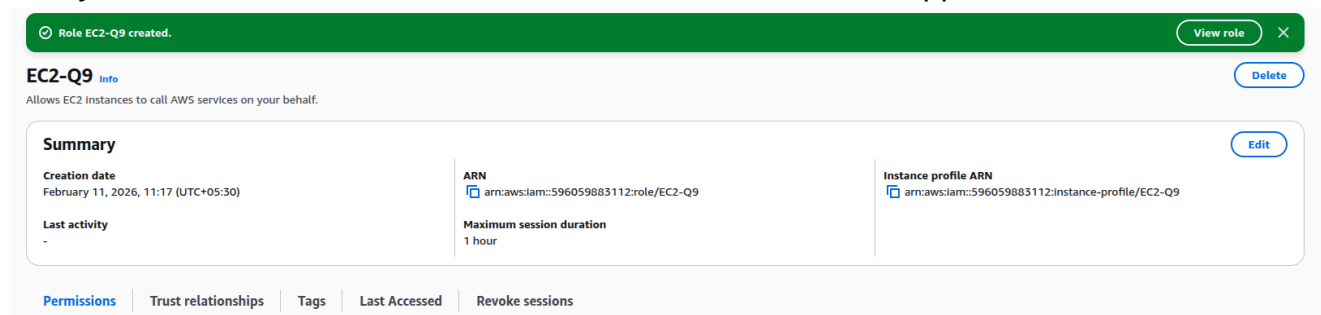
- AWS Key Management Service (KMS) – to decrypt encrypted configuration data and secrets.

- Amazon CloudWatch Logs – to push application and system logs for monitoring and troubleshooting

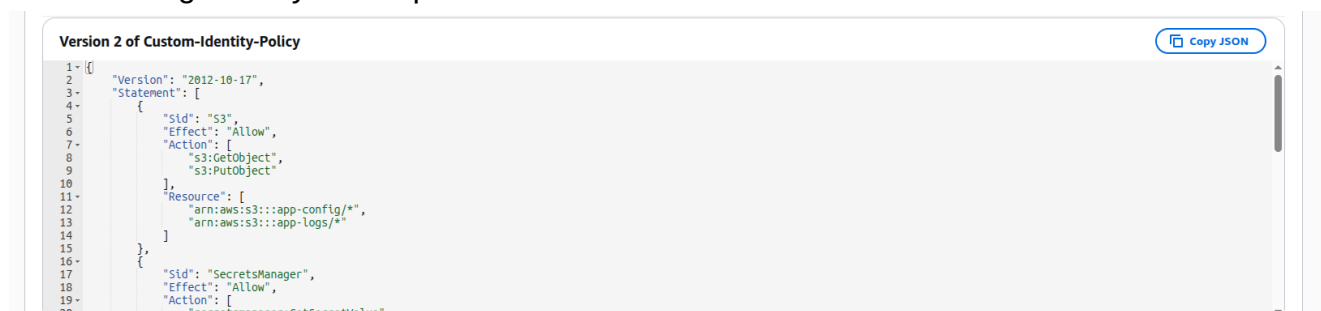
AWS RDS: To read data and write data from application to RDS.

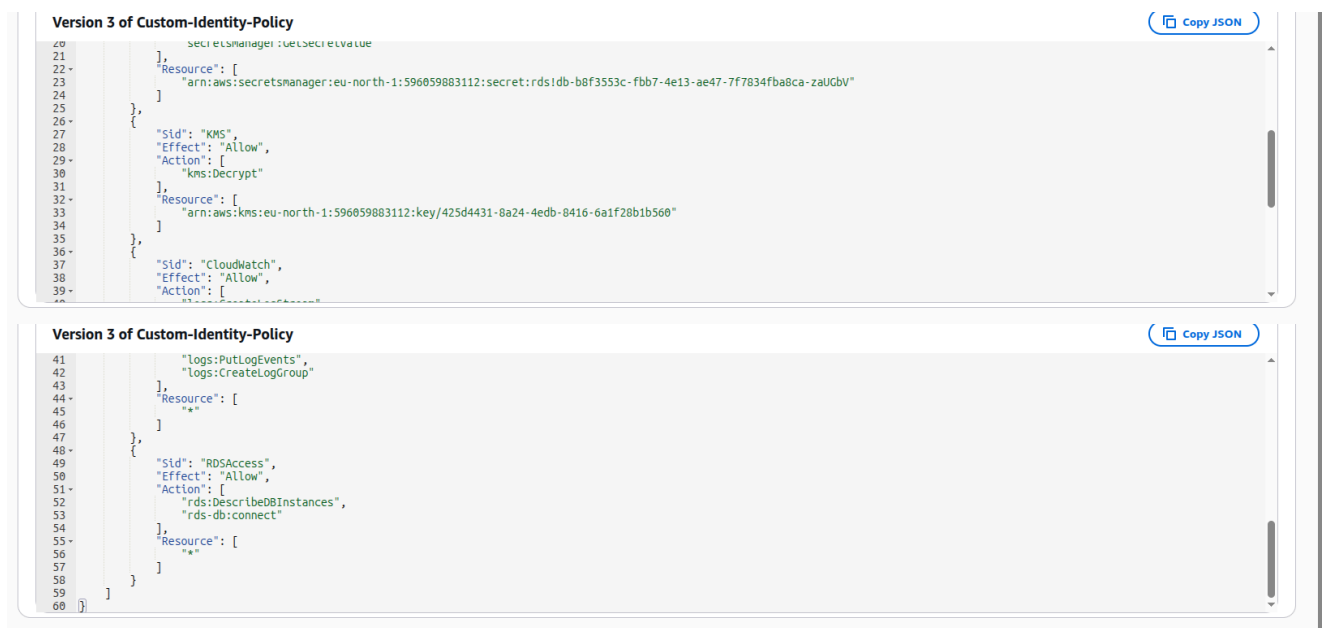
Ans:

Firstly I create an IAM role for the EC2 instance which runs the application .



Now creating Identity based policies for the resource ec2.

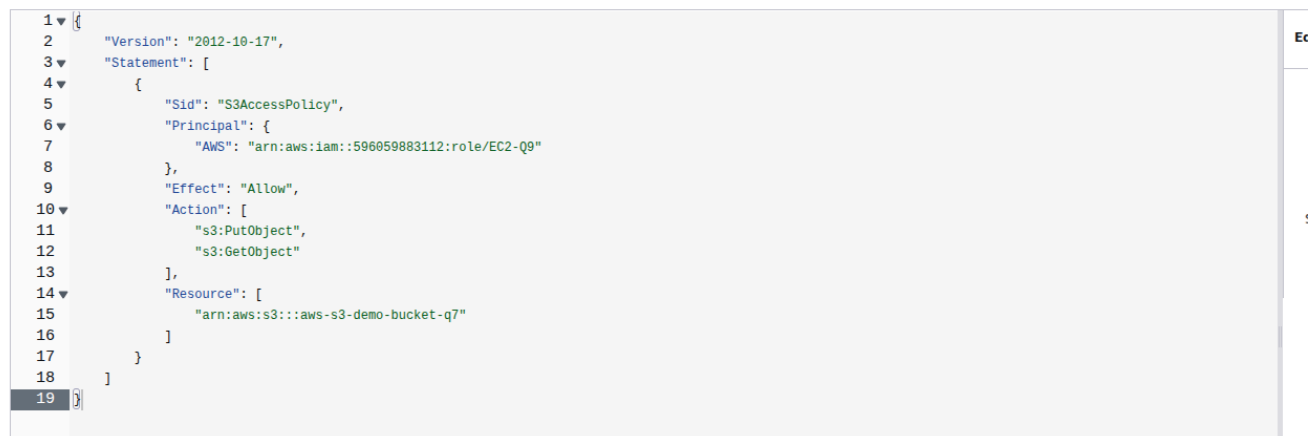




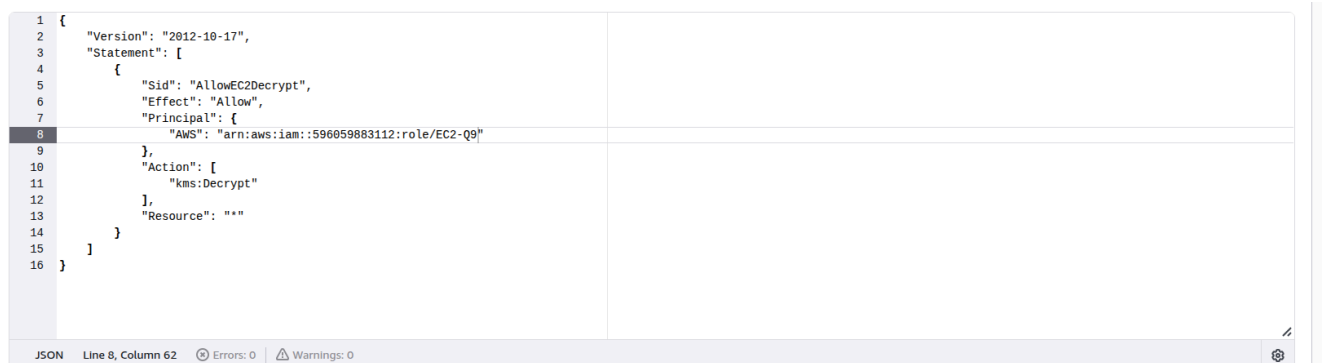
Now creating resource based policies for the resources to use the application in a secure way.

For S3 , I am using my bucket from Q7 only to attach a policy to it , that the role named EC2-Q9 which I created above will access the bucket to list and add object.

Policy



The policy for KMS



The policy attached to Secret Manager

Resource permissions - optional [Info](#)

[Edit permissions](#)

Add or edit a resource policy to access secrets across AWS accounts.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "AllowEC2Access",
6       "Effect": "Allow",
7       "Principal": {
8         "AWS": "arn:aws:iam::596059883112:role/EC2-Q9"
9       },
10      "Action": "secretsmanager:GetSecretValue",
11      "Resource": "*"
12    }
13  ]
14 }
15 }
```

Now verifying from the test ec2 to see i am only able to access the resources based on the permissions. I am checking it for my database instance . First I verify that role is attached or not.

```
[ec2-user@ip-10-0-0-173 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-173 ~]$ aws --version
aws-cli/2.32.22 Python/3.9.25 Linux/6.1.161-183.298.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-0-173 ~]$ aws sts get-caller-identity
{
  "UserId": "AROAYVVR7QAZUCDWCVCVAEG:I-0a96f94a3dae68db3",
  "Account": "596059883112",
  "Arn": "arn:aws:sts::596059883112:assumed-role/EC2-Q9/I-0a96f94a3dae68db3"
}
[ec2-user@ip-10-0-0-173 ~]$
```

Now I get the secret value first and then see the connection

```
aws <command> help
aws <command> <subcommand> help

[ec2-user@ip-10-0-0-173 ~]$ aws secretsmanager get-secret-value \
--secret-id rds\ldb-b8f3553c-fbb7-4e13-ae47-7f7834fba8ca \
--region eu-north-1
{
  "ARN": "arn:aws:secretsmanager:eu-north-1:596059883112:secret:rds\ldb-b8f3553c-fbb7-4e13-ae47-7f7834fba8ca-zaUgbV",
  "Name": "rds\ldb-b8f3553c-fbb7-4e13-ae47-7f7834fba8ca",
  "VersionId": "7b997244-1b13-4d94-b96e-9ace43cfeba9",
  "SecretString": "{\"username\":\"admin\",\"password\":\"vQ*-6F-)ItV[_3yuG#Kdh8!8?0Fw\"}",
  "VersionStages": [
    "AWSCURRENT",
    "AWSPENDING"
  ],
  "CreateDate": "2026-02-11T06:13:36.065000+00:00"
}
```

i-0a96f94a3dae68db3 (Test)

PublicIPs: 16.171.236.132 PrivateIPs: 10.0.0.173

[CloudShell](#) [Feedback](#) [Console Mobile App](#)

© 2025, Amazon Web Services, Inc. or its affiliates

The connection succeeded.

```
    ],
    "CreateDate": "2026-02-11T06:13:36.065000+00:00"
  ],
  "CreateDate": "2026-02-11T06:13:36.065000+00:00"
}
[ec2-user@ip-10-0-0-173 ~]$ mysql -h three-tier-mysql-db.czggwqayuo.eu-north-1.rds.amazonaws.com \
-u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 380
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| three_tier_db |
+-----+
5 rows in set (0.000 sec)

MySQL [(none)]> use three_tier_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [three_tier_db]> show tables;
+-----+
| Tables_in_three_tier_db |
+-----+
| users |
+-----+
1 row in set (0.002 sec)

MySQL [three_tier_db]> select * from users;
```

Q10.An EC2 instance uses an IAM instance role with *permissions to upload files to an S3 bucket, list bucket*

contents, read existing objects, and delete temporary objects. Explain how you would identify all AWS API operations (including implicit and failed calls) made by the EC2 instance role, distinguish between bucket-level and object-level permissions, and replace with a least-privilege IAM policy without impacting the application.

Ans:

The EC2 instance which uses an IAM role has permission to do everything with anything with the s3 bucket service.

The screenshot shows the AWS IAM console for a role named 'Q10'. The role's summary includes its creation date (February 11, 2026), ARN, and instance profile ARN. The 'Permissions' tab is active, showing a single policy named 'Q10Policy'. The policy's JSON content is displayed, showing a single statement that allows all actions on all S3 resources.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Effect": "Allow",
7       "Action": [
8         "s3:*"
9       ],
10      "Resource": [
11        "*"
12      ]
13    }
14  ]
15 }
```

To check all the AWS API operations (including implicit and failed calls) made by the ec2 instance role I will use the cloudtrail and add management event and data event to get all the API operations performed by it .

The screenshot shows the AWS CloudTrail console for a trail named 'piyush-cloudtrail'. The trail is enabled and logging is on. The details section shows the trail's location, log file validation status, and last log file delivered.

General details	Trail log location	Log file validation	SNS notification delivery
Trail logging Logging	aws-cloudtrail-logs-596059883112-80cee0d4/AWSLogs/596059883112	Enabled	Disabled
Trail name piyush-cloudtrail	Last log file delivered February 11, 2026, 22:15:31 (UTC+05:30)	Last file validation delivered February 11, 2026, 21:20:55 (UTC+05:30)	Last SNS notification -
Multi-region trail Yes	Log file SSE-KMS encryption Not enabled		
Apply trail to my organization Not enabled			

By using the cloudtrail logs and using the filter for the role we can get all the actions performed by it and then we can filter by eventName. Then we can analyse the logs further and then distinguish between the bucket-level permission and the object level permission by mapping the actions performed by the IAM role on the bucket . We can use the principle of least privilege to restrict our policy by replacing the * with more strict policy and action , by allowing only the action which are needed to be performed by the IAM role.